Bacteria are known to cause disease in plants, animals and Human beings. The List is given in Table 1.8, 1.9, 1.10 and Figure 1.15.

Table 1.8: Plant diseases caused by Bacteria

S.No.	Name of the Host	Name of the disease	Name of the pathogen
1	Rice	Bacterial blight	*Xanthomonas oryzae*
2	Apple	Fire blight	*Erwinia amylovora*
3	Carrot	Soft rot	*Erwinia caratovora*
4	Citrus	Citrus canker	*Xanthomonas citri*
5	Cotton	Angular leaf spot	*Xanthomonas malvacearum*
6	Potato	Ring rot	*Clavibacter michiganensis* subsp. *sepedonicus*
7	Potato	Scab	*Streptomyces scabies*

Table 1.9: Animal diseases caused by Bacteria

S. No	Name of the Animal	Name of the disease	Name of the pathogen
1.	Sheep	Anthrax	*Bacillus anthracis*
2.	Cattle	Brucellosis	*Brucella abortus*
3.	Cattle	Bovine tuberculosis	*Mycobacterium bovis*
4.	Cattle	Black leg	*Clostridium chauvoei*

Table 1.10: Human diseases caused by Bacteria

Serial No.	Name of the disease	Name of the pathogen
1.	Cholera	*Vibrio cholerae*
2.	Typhoid	*Salmonella typhi*
3.	Tuberculosis	*Mycobacterium tuberculosis*
4.	Leprosy	*Mycobacterium leprae*
5.	Pneumonia	*Diplococcus pneumoniae*
6.	Plague	*Yersinia pestis*
7.	Diphtheria	*Corynebacterium diptheriae*
8.	Tetanus	*Clostridium tetani*
9.	Food poisoning	*Clostridium botulinum*
10.	Syphilis	*Treponema pallidum*



(a) **(b)**

Figure 1.15: Plant diseases caused by bacteria (a) Citrus canker (b) Potato scab

Have you heard about the word "Probiotics"



Probiotic milk products and tooth paste are available in the market. *Lactobacillus and Bifidobacterium* are used to prepare probiotic yoghurt and tooth paste

Activity 1.3

Collect some root nodules of leguminous crops. Draw diagram. Wash it in tap water and prepare a smear by squeezing the content into a clean slide. Follow Gram staining method and identify the bacteria.



Bacteria forms Biofilms and leads to dental caries and Urinary tract infection (UTI)

Ralstonia synthesize PHB (Poly-β-hydroxyl butyrate) a microbial plastic which is biodegradable.

1.4.8 Archaebacteria

Archaebacteria are primitive prokaryotes and are adapted to thrive in extreme environments like hot springs, high salinity, low pH and so

11th Botany_EM_Chapter 1.indd 18 06/01/2023 19:44:54